CSE 299
RideRent Pro
One Hub, Easy Rental, Safe Journey

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***Abstract*—The RideRent Pro system is a smart vehicle rental platform that allows users to rent vehicles conveniently through a mobile application. The system provides online and manual booking facilities for different types of vehicles such as bikes, private cars, and microbuses. Users can view vehicle details, availability status, pricing, fuel type, ratings, and driver information before making a reservation. The platform integrates secure online payment methods including mobile banking and card-based transactions. RideRent Pro aims to improve accessibility, reduce rental complexity, and provide a seamless transportation experience for customers.**

***Index Terms*—Vehicle Rental System, Mobile Application, Online Booking, Smart Transportation, fleet Management, Driver Management,QR code Authentication,Customer Relationship Management,Vehicle availability Tracking,Real time Booking,Location Based Services,Cloud Based System,Payment Gateway, RideRent Pro**

I. Introduction

Transportation plays a significant role in modern society. Many people require temporary access to vehicles for personal, business, or travel purposes. Traditional vehicle rental services often involve lengthy paperwork, manual processes, and limited accessibility.

RideRent Pro is designed to simplify the vehicle rental process through a digital platform. Customers can browse available vehicles, compare prices, select drivers, and complete bookings through a mobile application. The system enhances user convenience while improving operational efficiency for rental service providers.

II. Problem Statement

The existing vehicle rental industry in Bangladesh faces several challenges, including limited access to reliable rental vehicles, lack of transparency regarding vehicle conditions, difficulty in finding trusted drivers, and inefficient communication between vehicle owners and customers. At the same time, many privately owned vehicles remain unused for significant periods, resulting in lost income opportunities for owners.

Most current rental services operate through traditional rental companies and do not provide a platform where individual vehicle owners can directly rent out their vehicles. Furthermore, customers often lack access to verified information about vehicle owners, drivers, and vehicle availability.

Therefore, there is a need for a secure and centralized digital platform that connects vehicle owners, customers, and drivers while ensuring transparency, trust, and convenience throughout the rental process.

III. Objective

The main objectives of the RideRent Pro system are:

*To develop a peer-to-peer vehicle rental platform. *To enable vehicle owners to list their vehicles for rent. *To provide real-time vehicle availability information. *To allow customers to browse and compare available vehicles. 8To provide detailed vehicle owner and driver information. *To integrate a secure in-app communication system. *To support vehicle rental with or without drivers. *To implement verification and rating systems for trust and security. *To create additional income opportunities for vehicle owners.

IV. Literature Review

The vehicle rental and ride-sharing industry has evolved significantly due to the growth of mobile applications and digital transportation services. In Bangladesh, platforms such as Pathao, Uber, and OBHAI have improved transportation accessibility through app-based ride-hailing services [1], [2], [3]. However, these platforms primarily focus on passenger transportation and do not provide centralized vehicle rental facilities or self-drive rental options.

Traditional vehicle rental agencies such as Garivara, Moriom Enterprise, and Sheba.xyz offer various rental vehicles, but most booking processes are still conducted through phone calls or messaging applications, resulting in limited automation and inefficient customer experiences [4].

Globally, peer-to-peer vehicle-sharing platforms such as Turo and Zoomcar have introduced innovative vehicle rental models. Turo enables vehicle owners to rent their personal vehicles directly to customers through a digital marketplace, while Zoomcar provides self-drive rental services and owner-based vehicle hosting programs [5], [6]. These platforms improve vehicle utilization and create additional income opportunities for vehicle owners.

Research has highlighted the importance of trust, identity verification, secure transactions, and driver authentication within modern transportation systems [7], [8]. Driver verification, customer reviews, vehicle ratings, and real-time availability management contribute significantly to customer confidence and service quality [9].

Despite these advancements, existing platforms generally lack a centralized hub-based vehicle management system, professional pre- and post-rental vehicle inspections, AI-powered trip assistance, and integrated reward mechanisms for customer retention [5], [6]. Furthermore, many solutions are not specifically designed to address the operational and security challenges of developing countries.

RideRent Pro aims to address these limitations by providing a centralized hub-based vehicle rental ecosystem that includes vehicle owner verification, driver portfolios, mandatory vehicle inspection procedures, AI-assisted booking support, customer review systems, and flexible rental options with or without drivers.

V. Proposed System

RideRent Pro is a mobile-based peer-to-peer vehicle rental platform that connects vehicle owners, customers, and drivers within a single digital ecosystem.

Vehicle owners can register their accounts, verify their identities, and list their vehicles by providing vehicle details, rental pricing, available dates, and time slots. Customers can browse available vehicles, view vehicle specifications, owner profiles, ratings, reviews, and availability schedules before making rental decisions.

The platform also provides verified driver profiles containing driver information, experience, ratings, and customer reviews. Customers can choose to rent a vehicle either with a driver or without a driver according to their requirements.

A built-in messaging system allows direct communication between customers and vehicle owners, enabling discussions regarding rental terms and availability. The platform includes verification mechanisms, review systems, and administrative monitoring to ensure transparency and trust among all users.

VI. Methodology

The development methodology of RideRent Pro follows a structured workflow.

1.Vehicle owners register and complete identity verification. 2.Owners add vehicle information, rental price, and available time slots. 3.Customers register and browse available vehicles. 4.Customers view vehicle details, owner profiles, ratings, and reviews. 5.Customers may also browse available driver profiles. 6.Customers select a vehicle and choose whether a driver is required. 7.Customers communicate with vehicle owners through the in-app chat system. 8.A booking request is submitted to the vehicle owner. 9.The owner accepts or rejects the booking request. 10.Upon approval, booking confirmation is generated. 11.After rental completion, customers can provide ratings and reviews for both the vehicle and the driver.

VII. System Features

The RideRent Pro system provides the following features:

- User Registration and Login
- Vehicle Owner Verification
- Vehicle Listing Management
- Vehicle Availability Calendar
- Vehicle Search and Filtering
- Vehicle Details and Rating Display
- Management of the Driver Portfolio
- Driver Selection Facility
- In-App Customer-Owner Chat System
- Management of reservation requests;
- Customer Reviews and Ratings
- Driver Reviews and Ratings
- Owner Reviews and Ratings
- Secure User Verification System
- Admin Monitoring and Fraud Prevention

VIII. Expected Outcomes

The implementation of RideRent Pro is expected to create a reliable peer-to-peer vehicle rental ecosystem that benefits both vehicle owners and customers. Vehicle owners will gain additional income opportunities by renting out their unused vehicles, while customers will have access to a wider variety of rental options.

The system is expected to improve transparency, trust, and convenience through verified profiles, rating systems, driver portfolios, and secure communication channels. Ultimately, the platform will contribute to better vehicle utilization and support the digital transformation of transportation services.

IX. Future Work

Future enhancements of RideRent Pro may include:

* GPS-based vehicle tracking item. * AI-powered vehicle recommendation systems. * Dynamic pricing based on demand and availability. * Online payment gateway integration. * Vehicle insurance management. * Automated fraud detection mechanisms. * Emergency assistance services. * Real-time location sharing during rentals. * Electric vehicle rental support. * Predictive analytics for rental demand forecasting.

X. Conclusion

RideRent Pro is a smart peer-to-peer vehicle rental platform designed to connect vehicle owners, customers, and drivers through a secure and user-friendly mobile application. The system promotes efficient vehicle utilization by enabling owners to rent out their vehicles while providing customers with flexible transportation options.

By integrating verified user profiles, driver portfolios, vehicle ratings, customer reviews, and real-time communication features, the platform establishes a trustworthy and transparent rental environment. RideRent Pro has the potential to transform traditional vehicle rental services in Bangladesh by introducing a modern sharing-economy model that benefits all stakeholders while supporting the future growth of smart mobility solutions.

XI. REFERENCES

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